

ActInf GuestStream 075.1 ~ Matsumura et al.: "Active Inference With Empathy Mechanism"

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hello and welcome it's March 13th 2024 and we're here in active inference guest stream number 75.1 active inference with empathy mechanism for socially behaved artificial agent in diverse situations with t Yuki matsumura kanako isaki and huki Mizuno so thank you all for joining and looking forward to hearing your talk and discussion okay thank you for inviting us to such a good opportunity and I'm T and my co researcher kanako and huki also join this discussion we are developing AI especially AI which is inspired by humanik autonomous agent to make social artificial agent recently we published a paper related to the idea in Journal of a life the title is active imp with empathy mechanism for socially behaved artificial agent in diverse situations first I'd like to explain the paper this is the motivation of This research at the title of the paper shows the motivation of This research is to develop socially behaved artificial agent we believed that sociality will become an important point to apply AI in our daily life the difficulty to realize social agent is that an appropriate social behavior depend on a given situation or culture as described in the bottom figure we have sociality even in just working case for example how fast should we walk or how much distance from others orah blah blah so if we Implement sociality with rule based program we have to design a lot of magic number for each situations this is impossible because there are a white variety of situations due to this unified mechanism or principle for social behavior is required this is the challenge we assume that we humans has such unified behavior model because we humans can behave socially in diverse situations and we develop social Eng inspired by human behavior model namely three principle three energy principle and active inflence in this research we use the active inflence for the basis of the behavior model of human life agent as you know active inference is a concept proposed in the free energy principle and free energy principle is a unified hypo for our variety of cognitive activities in the FP our cognitive activities are explained by such that first we have internal models for predicting future of surrounding environment and second we always try to minimize the free energy of the model

free energy is defined by this equation and it intuitively represent uncertainty of prediction the interesting point of the FP is that the internal model if not only the hidden state but also actions for the agent that is the action to be taken is the action which minimize the free energy of the agent this is the basic idea of active inference as I understand it intentional actions also can be explained in the active inference in this case we consider expected free energy which is the free energy for the future State when considering much time Step Ahead expected energy at much time Step Ahead is considered as described in the right figure based on the expected free energy for each action the distribution the action is determined such that the smaller the expected free energy the higher the probability of selection when we consider intentional Behavior intentions are considered as PRS of future observations and reward according to the Future observation are encoded into the first time into the uh in the last equation these are my understanding of the standard active inference and the behavior model of the humanik agent based on it we extend this active inference to generate the social behavior the idea of the extension is explained using a situation in which there are two agents called I and other the agent called I is the subject of the action in this explanation the biological agent under the AC inference makes predictions about the environment and take some action to minimize the uncertainty of the prediction in this figure I predict future beh I predict future behavior of the other because the a is an environment for I namely I act to reduce uncertainty for the for the other so far this is a standard active inference here if we consider that the other is also a biological agent we can assume that the other also act under the active inference under this idea we can assume that the other also predict the future of behavior of me and the other tr to is the uncertainty of the prediction about me in this session in this understanding we notice that our action can reduce not only our own free energy but also that of the other this is because my action affects not only my own free energy but also the free energy of the others if I can predict others prediction for me I can act to I can act to release the free energy of the other this is the idea of the proposed extension of the active inference we extend FP and active inference from an individual idea to an Collective idea as described in this image of concept the agent imaginary share the brain and they try to minimize the free energy of the shared brain to relas the free energy for the shared brain we act not only for me but also for the others because of this action for others we expect that this idea can be used to generate social behavior which is a behavior for us not only only for me or only for you to realize this idea it is necessary to predict others prediction about myself this is implemented based on the mirror system and simulation Theory hypothesis for the mechanism of empathy in this in the miror

system and simulation Theory it is that that we understand others by simulating others by own bodies we consider the idea of using our own body to simulate others as using our internal model to infer other prediction the overview of the mechanism is described in the figure the agent has the internal model to predict future observation for his own observation moreover we use the same internal model to predict others prediction about myself At the model is the same the input to the model is changed to the observation of the others for this the process of inference of others observation is included for example when the observ is an image Noel view synthesis which is the problem of generating images from different Viewpoint can be used in this in the other case as we will discuss in the following evaluation if the observation is the location of the agent for ordinate transformation transformation can be used based on this inference procedures the expected free energy is is expended as described in this equation as with the encoding of intentional Behavior we add a time to the first time that is intentional to respond to the expect expectation of others if there are multiple others the some of each other expectation is considered we call the proposed idea empathic activ we evaluate the proposed idea using the control task of an autonomous autonomous mobile agent in the evaluation multiple agent move to each Target point the shortest path of each agent intersect each other and agent are required to take cision avoidance behavior the prayer this red agent is controlled by the proposed method and the other agent are controlled by the social force model social force model is a simple social working model roughly set it is a Dynamics model in which the walking speed and direction are determined by two forces the first one is driving force to the go and the second is repulsive force from the others in this evaluation the weight of the weight the weight of these two type of forces are changed to evaluate different situations where are two type of others selfish or aristic others when the for to the goal is larger than the force from others the agent take behavior in selfish on the other hand the force for the from the others is larger than the PO to the go the agent take behavior in alistic this a example of the behavior of others under the control of the social force model the player let CLE randomly moved and the others green blue or orange moved with social force model the top picture is an example for selfish others and the bottom picture is an example for an aristic a the difference is from the difference of the magnitude of the two forces as you can see from the figures when other are characteristic the other will take the p with uh P which has more margin with others the setup for the prayer is simply disre described the action space is defined in discrete space movement for five directions and weight weight current position totally six action is assumed a simple variational o encod like model is used to the internal model and the model is with the offline

man the model is used to predict the expected free energy for each action first the evaluation result when there is only one other besides the prayer as shown we discussed the poity of the agent based on the for view point the first one if my travel my travel distance is longer it is social behavior and the second if the other travel distance is shorter it is social behavior the start if the minimum margin among agent is larger it is social behavior for if the inequality of the travel distance is smaller it is social behavior inequality means that the difference in Trav distance among agents as a comparison we also show the result of the standard active inference cases the two figures in the left side are the case when the others are selfish in this case the prayer with the standard active inference moves straight to the destination because the prayer Act optimally only for its own purpose on the other hand the proposed agent to Appo the others as well as as well as the other because the prayer predict that the other expect to the prayer to avoid collision and the prayer try to respond to it as a result of taking this the two P the sociality of the behavior is increased namely the travel distance of the prayer increased compared to the that of the standard active inference case at the same time the travel distance of the others decrease and the minimum distance between the prayer and the other increase in addition the inequality of the difference in travel distance is decreased next the case where other aristic is shown on the light two figures in this case the same tendency is observed as in the case where of others other is selfish but the player moved a pass with larger margin to the other to the other compared to the case of selfish other the result indicates that the proposed agent can behave in Social manner that is that beneficial to the other even if the result are disadvantageous to itself an important point is that the difference of the social behavior does not depend on any reward designing as in reinforcement running namely the sociality depend on the character of the other around the prayer the other around the prayer is not just obstacle for the agent but uh as a a teacher for the social [Music] manner s in the case where three others besides the prayer are similar result observe in the aristic as a case namely the travel distance of the prayer is increased and that of others are decreased and the minimum margin is increased and inequality of the travel distance is decreased on the other hand in the case of selfish others a collision occurred it may be due to high density in the center point in this case the proposed agent can avoid collision by waiting the crossing of others I repeat the important point there is no need to design leward or penalty for sociality in each situation a social behavior realized by respon responding to the other expectation finally I would like to summarize the explanation of the paper as I said at the first SL the motivation of This research is to develop unified mechanism for social behavior the answer for the motivation is that

learning an internal model to predict the future behavior of others this is same as standard free energy principle or inference more more over predict others expectation about me with the same internal model and try to respond to the others expectation this this process make a cyclic process and it encourage the agent to behave like surrounding others because due to mirror system other expectation to me are same same to my expectation to others this mechanism encourages agent to behave like surrounding others this summary shows our idea for what is social behavior the answer is that social behavior is a behavior like surrounding others the others are not obstacle but teacher for social behavior this also mean that we are teacher of AI or robot for the future Society that's so this is the explanation of the paper i' like to discuss about it thank you would either of the other authors like to say hello or to give any remarks hello I amaki also a researcher at and um um I I have been studying on the control of system especially system with hardware and uh my recent interest is how to apply the after robot system hello my name is H I'm responsible for This research thank you for giving us this wonderful time thank you okay many interesting um questions but definitely very striking results about the tolerance to a variety of different social settings for the active inference algorithm to by nature of always updating at each time step based upon the expectations able to kind of deal with a range of different settings like different numbers of Agents and the different um Behavior types for the agents um I'll I'll ask some questions that were submitted by hear you Sam no no or okay now okay all good all good okay I will read some questions that were submitted by email by shohi wakayama and also we'll look to the live chat so I'll start with the fourth question he wrote I would like to receive more detailed explanations of no operation action the noop in general because I noticed also in the graphics that when there's going to be a collision that the active inference agent just stops so how did you model or calibrate that decision to make a move at all versus which way to move forward okay first I'd like to answer what is the no operation a NOP I mean no is just a no operation so in the evaluation case not means wait the current po the agent take no movement action the agent stay the current position this is a not operation no operation and the question is very important because this is a critical problem to realize our method the problem is that we want to decide an action but the procedure we proposed requires to decide action as described in this figure the let Circle action actions for others we decide to action of the agent but it requires the deciding other action this cyclic requirement is very difficult problem so we have to set some option for the others the S I think there is there are solution straty three solution strategies the first one is assume random action apprising the result of the randomly selected actions the

second is predict action learning and predict using policy Network for example Alpha go use this strategy and the last one is we selected the third strategy is that assume specific action no operation means I don't take any actions and you act as a priority please I mean NOP I mean this idea you for assuming no so I believe that humans has such sociality in nature this is the reason why I use the sad idea does it make sense yes okay next um question how do you tune or calibrate the model so that it's in an area of State space that's pro-social like maybe you could have a generative model that just only did nope and it was just paralyzed or another one just goes straightforward and no matter what so how do you explore that state space and choose which parameters are open for updating and which parameters are going to be like explored during the design phase and then fixed at runtime oh at runtime uh I don't adjust any parameter at long time the model is Tred and offline Manner and the parameter model is explore explore searched at the training phase but the parameter space is not so huge in this case because the model is very simple the observation is just a two dimensional information X and Y that means location of the agent so in the case of four agent there is just eight dimensional space but we need sometime steps but even that case the dimension is not so large so the hidden state is also not large can you understand yes evaluation situation is very simple so the model is also simple so I don't search parameter with a lot of IND that is not PA simply just to try to some parameter and I pick up a best parameter set so this was a this was an in silico software only simulation right oh yes this is just the simulation about simulation so um how did you tune it and then I guess a bigger question and definitely one that a lot of people are interested in in learning active inference is like how do you go from having a in silico simulation to bringing that onto a physical robot like what are the steps in the process between having the situation in the simulation and bringing it into like a room yes that's a very interesting point and it's very important and Han try to app the active inference into the real robot so how about do you think kind um so uh so I I I I also tried in a small uh space for to apply the this after to the robot so um I have just an idea that um um maybe uh The Continuous space is the important um key for to uh PRS simulation to the physical world and in general uh the problem is also the problem in reinforcement learning I think so the techniques in reinforcement learning field also can be applied into the active inference cases for example simulation to simulation to real I I don't remember the name of technique but s maybe such techniques can be applied in the case of active inference but I don't know there is a active inference specific technique I don't know such Tech that's interesting about the continuous State spaces because in the discrete State space where you have the markov decision

process you get the big planning trees but our experience like walking through a crowd or maybe waiting at a intersection it's not necessarily based upon planning it's a lot more on the path and the curvature of the path that's planned and maybe the paths that others are expected to take as they move through space so that's possibly like a simpler representation yes that could give natural movement Behavior with a just a few parameters thank you and there is a another problem for applying this technique to a real world that is a computational cost because we use the monard simulation to start the best option I think this is not sweet to to the real situation so if we try to apply this technique to the real application we have to solve the computational cost problems I think so yeah another part even outside of the decision making computational costs would be like going from the video camera data oh yes when you're getting the exact location data perfectly you're in the minimum you're like kind of playing chess in the state Space versus to have noisy sensors there might need to be a lot of um uh observation model here a lot of layers before the simple kind of active INF could even happen okay I'll ask another question from uh shohi so question five he wrote in this study the behaviors of the standard active inference agent and the empathetic agent are compared I would like to know why you did not compare the proposed method with other game theoretic approaches because the metrics like travel distance can be appli appable to these methods as well oh the reason why I compare the proposed method with active inference is that the aim of the paper is to extend the active inference to the social social behavior manner so it is not I think it is natural to compare the proposed me standard activ forance but the question is that why we don't compare the proposed method with the other games theoretic method I don't I'm not sure what this means the games theoretic method but if the that mean the reinforcement running method the reason why I don't compare it is that just it need a lot of time to compare it I I think it is important but the main focus of this paper is to evaluate quantitative future of the propos method and the qualitative varation is not so important for us at the moment but I think it is important and I have to compare it in future work so there is no real reason this is just uh we don't have uh much time to compare it thank you makes sense um another question so the movement was being influenced by the relative ratio of the F other and F goal yes so here F other is kind of like a um electrostatic repulsion from other agents how does it how does it come into play that the red agent is entertaining beliefs about what other agents think sorry can you repeat the question yeah so here the two forces acting on the red agent are the the gravity for the goal and the repulsion for the proximity of other agents yes so where or how does it come into to play that the agent is actually thinking what other agents are going to do like the

birds flocking simulations that have the simple rules like go in the same direction as the other birds and maintain a distance between them but that doesn't require a theory of mind for the other birds Oh you mean how the agent know the H from the others or ah you mean how the agent know the course of goal of others yeah it's just running away from others or is it actually imagining what they so uh that's good question yes uh in this simulation uh s the answer is that the destination destination the others given I don't remember the detail of the evaluation I but a Sor the agent don't have to know the information at first I think I give the information to the agent but that information does not need the asan yeah so the goals of the others are not not expressly given but uh the with internal model uh the agent can predict um the path test so we can pred uh go like go but uh not not exped does it exp does the red agent explicitly model what the others are do or it just has heris stics that it follows when they move the foral one so only predict the action recent action and yes near future yes the agent just learn the future position of the others this is the all of the task of the internal model so there's no need to give the go of the others yes I think this is very interesting aspect like to what extent the social behavior rests upon Simple Rules like etiquette like passing on the right or on the left that don't require knowledge about the internal states of others or even what kind of thing they are whereas the experience of sociality and a lot of the psychology of Sociology focuses on very cognitive mechanisms that are very expressive you this is because the action is repeated to decided in short time duration so this is uh future of the proposed method as you said thank you I don't notice that that is um where are you going to go in your continuing research or development directions sorry future please uh I think uh an important point is the regulating the empathy and we have a presentation the last year uh conference about artificial life in the presentation we discussed how regulates the emotion our empathy this mean that we to uh the paper we in is that the expected the pre energy is extended to the including others so D called D my plus the others and the there is there should be weight of empathy that means W in this equation we have to regulate parameter I don't consider the parameter in the Journal paper but we have to regulate this parameter because there's diverse others there's good person or there's part person we should not empathy to the part address so this is the important point I'll emphasize others and we tackle this problem with the emotion emo we tled this problem with the regard to the emotional context okay this is a a future work and we publish uh we have a presentation at the at the current conference so if you have some interesting please read this paper that's very interesting like in a lot of the active inference and social sciences conversations that we've had here there's a lot of discussion about like how do you balance the

the coefficients in the free energy calculations for self and others yes because you're making decisions at a very short time scale like speech and second by second but then sociality is very longterm and so you could make a decision that is um aligned locally but then very out of alignment at a longer time scale especially if these other four have complex relation relationships then action is not very simple Yes is there any other um things that the other authors want to say or anything else you want to talk about I I don't have cool well thank you very much for joining the discussion thank you thank you yes good luck with the ongoing work and I hope to stay in touch and hear more about the research in the future thank very much okay thank you see you later bye fine for